# Walks, Paths and Connectivity from the Adjacency Matrix: A Study Note on $(A^k)_{ij}$ and the Connectivity Test

Study note prepared from a lecture whiteboard and the course textbook

**Abstract**

A graph with $N$ vertices can be stored as an $N \times N$ table of zeros and ones, its adjacency matrix $A$. This note answers three questions about that table. First, the entry in row $i$ and column $j$ of the matrix power $A^k$ is the number of walks of length $k$ from vertex $n_i$ to vertex $n_j$; it counts walks, not paths [2, 1]. Second, a graph with $N \geq 2$ vertices is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive, or equivalently if and only if $(I + A)^{N-1}$ has no zero entry. Third, a path between $n_i$ and $n_j$ is a walk from $n_i$ to $n_j$ with no repeated vertex, which is the course textbook's simple path. Both theorems are proved in full, because the course textbook does not state them in the pages checked [9]. Every matrix in the examples was computed by script.

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# 1 Introduction

## 1.1 The question and where it came from

A lecture whiteboard shows a small graph, its table of zeros and ones, and two questions about that table [4, 5]. The board states the problem as follows: a graph with $N$ nodes is given through its adjacency matrix $A$; (1) what does the entry in row $i$ and column $j$ of a power of $A$ represent, and (2) how does one know whether the graph is connected [4, 5]. The board also gives a definition of a connected graph, which uses the word "path" [4, 5]. The assignment text added a third question: what is the definition of a path between two vertices $n_i$ and $n_j$. That third question comes from the assignment, not from the board. The paper therefore answers three questions.

Q1. What does the entry $(A^k)_{ij}$ of the $k$-th power of the adjacency matrix represent?

Q2. How can one decide from $A$ alone whether the graph is connected?

Q3. What is the definition of a path between the vertices $n_i$ and $n_j$?

*Reading of the board.* In question 1 the board puts a small mark above the letter $A$. In both photographs the mark looks like a small $n$ or a pair of quotation marks, not clearly a $k$ [4, 5]. This paper reads the mark as an exponent $k$, because the assignment text asks about $A^k$. This is an interpretation, and it is qualified: the board does not settle it. It does not change the mathematics, because Theorem 3.5 holds for every positive whole exponent, whatever letter is used.

### 1.2 The story used throughout

Every explanation marked "In plain words" uses one story. A village has a number of houses. Each house is a point. Each road joins exactly two different houses. Every road can be driven in both directions. No road leads from a house back to itself. No two roads join the same pair of houses. A trip is a list of houses, one after another, where each house is at the other end of a road from the house before. A trip may pass the same house twice and may drive the same road twice. A clean route is a trip that never visits the same house twice. The village of the board has three houses in a row: a road from the first house to the second, and a road from the second house to the third. The first and third houses have no road between them. The mathematics below uses the same village; the terms are matched in Table 1.

Table 1: The story and the mathematics. The right column names the definitions that are introduced in Section 2.

| Story | Mathematics |
|---|---|
| house | vertex |
| road | edge |
| village | simple undirected graph (Definition 2.1) |
| trip | walk (Definition 2.18) |
| clean route | path (Definition 2.31) |
| table of roads | adjacency matrix (Definition 2.13) |
| village in one piece | connected graph (Definition 4.1) |

### 1.3 What the paper shows

The answers, stated briefly and proved later, are these.

- *Q1* (Section 3). For $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length $k$ from $n_i$ to $n_j$. The statement appears in standard lecture notes [2, 1]; the proof in this paper is written out in full. The entry counts walks, not paths, and Proposition 3.11 gives a counterexample. One web source states the theorem with the word "paths", which is wrong for the usual meanings of that word [7]; this is discussed in Section 7.3.
- *Q2* (Section 4). For $N \geq 2$ vertices the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive. A published note states this test and lecture notes state its second form, and both prove it loosely [7, 1]. The proof in this paper is complete (derived here).
- *Q3* (Section 5). A path between $n_i$ and $n_j$ is a walk from $n_i$ to $n_j$ with no repeated vertex. This is the course textbook's simple path, Definition 1.2.13 [9, p. 22].

The course textbook states neither the walk-counting theorem nor the matrix test in the pages that were checked, and a search of the whole book found no such theorem or test; the absence is qualified because the rest of the book was not read page by page [9]. Both results are therefore proved here, and the text says so each time.

### 1.4 Conventions, assumptions, and kinds of statement

**Assumptions.** The graph is finite, simple, and undirected. This is the setting of the assignment, and it agrees with the symmetric matrix with zero diagonal on the board [4, 5]. The textbook's graphs are of this kind too [9, p. 9]. Directed graphs and graphs with repeated edges are not treated (Section 7.5).

**The example graph.** The board shows a drawing of three vertices and, beside it, a matrix with rows (0 1 0), (1 0 1), (0 1 0). The matrix encodes the two edges $n_1n_2$ and $n_2n_3$ only. The drawing shows three lines: one from 1 to 2 with an arrowhead at 2, one from 1 to 3, and one from 2 to 3 [4, 5]. So the drawing has an edge 1–3 that the matrix lacks, and an arrowhead that a symmetric matrix does not support. The problem statement says that the graph is given through its adjacency matrix [4, 5]. This paper therefore takes the matrix as the example graph, calls it the *board graph*, and treats the extra line and the arrowhead as a mismatch that it reports and does not resolve. When a graph with a cycle is needed, the drawing's triangle is used and is labelled as the triangle.

**Notation.** Table 2 fixes the letters. The board writes $(G, V)$ for the graph and $e_{ij} \in V$ in its matrix definition, where its $V$ is the edge set; the textbook writes $G = (V, E)$ with $E$ the edge set [4, 5, 9]. This is a difference of notation only. The board names the vertices $n_i$; the textbook names them $v_i$ [4, 5, 9]. This paper uses the board's $n_i$.

Table 2: Notation used in this paper.

| Symbol | Meaning |
|---|---|
| $G = (V, E)$ | a finite simple undirected graph with vertex set $V$ and edge set $E$ (textbook letters) |
| $N$ | the number of vertices, $N = \|V\|$ (the board writes $\|G\| = N$) |
| $n_1, \ldots, n_N$ | the vertices (the board's names; the textbook's $v_i$) |
| $uv$, $e_{ij}$ | the edge between $u$ and $v$; the edge between $n_i$ and $n_j$ |
| $A$, $A_{ij}$ | the adjacency matrix and its entry in row $i$, column $j$ |
| $A^k$, $(A^k)_{ij}$ | the $k$-th power of $A$ and its entry in row $i$, column $j$ |
| $I$ | the $N \times N$ identity matrix (ones on the diagonal, zeros elsewhere) |
| $k$ | the length of a walk, or an exponent; a positive whole number unless stated |
| $S$ | the sum $A + A^2 + \cdots + A^{N-1}$ (Definition 4.9) |
| $\deg(n_i)$ | the degree of $n_i$ (Definition 2.9) |
| $d(u, v)$ | the distance between $u$ and $v$ (Definition 2.41) |

**The word "path".** The word has more than one meaning in the sources. The textbook's own "path" (Definition 1.2.12) allows a repeated vertex, and it calls a path without a repeated vertex a "simple path" (Definition 1.2.13) [9, p. 22]. Two English texts call the second kind a path and the first kind a trail [3, 8]. The assignment states that in the course a path has no repeated vertex; that statement comes from the assignment text and not from a file, so it is recorded as an interpretation. This paper uses *path* for a walk without repeated vertices and *trail* for the textbook's Definition 1.2.12, and Table 3 gives the three conventions side by side.

**Kinds of statement.** A statement with a citation is taken from the cited source. A statement marked *derived* is proved or computed in this paper; every number in such a statement was computed by script (Appendix D). Statements marked *interpretation*, *estimate*, or *open* are

labelled in the same way, and the textbook's definition numbers are always written "textbook Definition 1.2.10" and so on, so that they cannot be mistaken for the numbers of this paper.

### 1.5 Organization

Section 2 builds the prerequisites in the textbook's order: graph, adjacency, the adjacency matrix, walk, trail, path, cycle, and distance, with the lemma that every walk contains a path. Section 3 answers Q1, Section 4 answers Q2, and Section 5 answers Q3. Section 6 works the board graph and three further graphs through in full. Section 7 states the limits, and Section 8 gives the direct answers. The appendices hold longer derivations, the textbook's terms, the transcription of the photographs, and the scripts.

## 2 Background: graphs, walks, and paths

Throughout, let $G = (V, E)$ be a finite simple undirected graph with $V = \{n_1, \ldots, n_N\}$, let $u, v$ be vertices and let $e, e'$ be edges, as fixed in Table 2. The definitions follow the textbook's build order [9]. Where the textbook has a number for a definition, the number is quoted as "textbook Definition $x.y.z$". After each definition come a remark on what it allows and forbids, an example, and a non-example.

### 2.1 Graphs, adjacency, and degree

**In plain words.** *A village has a list of houses and a list of roads. A road always joins two different houses. No two roads join the same pair of houses. Roads have no direction. The village is small enough that we can count its houses.*

**Definition 2.1** (Simple undirected graph; textbook Definition 1.1.1)**.** A *simple undirected graph* is a pair $G = (V, E)$ where $V$ is a finite nonempty set, whose elements are the *vertices*, and $E$ is a set of unordered pairs of distinct elements of $V$, the *edges* [9, p. 9].

**Remark 2.2.** The definition forbids a loop, because an edge is a pair of distinct vertices. It forbids a repeated edge, because $E$ is a set, so a pair occurs at most once. It forbids direction, because the pair is unordered, so $n_i n_j$ and $n_j n_i$ are the same edge. It allows a vertex that lies on no edge, and any finite number of vertices.

**Example 2.3.** $V = \{n_1, n_2, n_3\}$ and $E = \{n_1 n_2,\ n_2 n_3\}$. This is the board graph: three houses in a row.

**Non-example 2.4.** Two vertices $n_1, n_2$ joined by two roads, or a road from $n_1$ to $n_1$. The first repeats an edge and the second is a loop, so neither is a simple graph.

**In plain words.** *Two houses are neighbours when one road joins them directly. Being neighbours says nothing about longer trips.*

**Definition 2.5** (Adjacent vertices; textbook Definition 1.1.4)**.** Vertices $u$ and $v$ are *adjacent* if $uv \in E$; then $u$ is a *neighbour* of $v$ [9, p. 11].

**Remark 2.6.** Adjacency is symmetric, and no vertex is adjacent to itself, because there are no loops. It describes direct roads only, not longer connections.

**Example 2.7.** $n_1$ and $n_2$ are adjacent in the board graph.

**Non-example 2.8.** $n_1$ and $n_3$ are not adjacent in the board graph, although the trip $n_1, n_2, n_3$ joins them.

**In plain words.** *The degree of a house is the number of roads that touch it. A busy house has a large degree.*

**Definition 2.9** (Degree)**.** The *degree* $\deg(n_i)$ of a vertex $n_i$ is the number of edges that contain $n_i$.

This is the usual meaning of the word, stated for example as "the number of edges at that vertex, or the vertex degree" [2]. The textbook's own definition of degree was not read, so no textbook number is given.

**Remark 2.10.** In a simple graph the degree is also the number of neighbours, because each neighbour is joined by exactly one edge.

**Example 2.11.** In the board graph $\deg(n_1) = 1$, $\deg(n_2) = 2$, and $\deg(n_3) = 1$.

**Non-example 2.12.** $\deg(n_1)$ is not 2 in the board graph, although two other vertices can be reached from $n_1$. Only roads at $n_1$ count, and $n_3$ is reached by passing through $n_2$.

## 2.2 The adjacency matrix

**In plain words.** *Make a square table with one row and one column for every house. Write a one where the row house and the column house share a road. Write a zero everywhere else. This table describes the whole village.*

A *matrix* is a rectangular table of numbers; $A_{ij}$ is the number in row $i$ and column $j$. The *transpose* of a matrix swaps its rows and columns, and a matrix is *symmetric* if it equals its transpose, that is, if $A_{ij} = A_{ji}$ for all $i, j$.

**Definition 2.13** (Adjacency matrix; textbook, printed p. 11, after Definition 1.1.6)**.** The *adjacency matrix* of $G$ is the $N \times N$ matrix $A$ with $A_{ij} = 1$ if $n_i$ and $n_j$ are adjacent, and $A_{ij} = 0$ otherwise [9, p. 11].

The textbook's definition number for this matrix was not recorded, so only its page is given. The board writes the same rule as $A_{ij} = 1$ if $e_{ij} \in V$ and $A_{ij} = 0$ if $e_{ij} \notin V$, with its $V$ standing for the edge set [4, 5].

**Remark 2.14.** The definition allows only the entries 0 and 1. It depends on how the vertices are numbered: renumbering permutes rows and columns together.

**Example 2.15.** The board matrix $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ is the adjacency matrix of the board graph of Example 2.3: $A_{12} = A_{21} = 1$, $A_{23} = A_{32} = 1$, and every other entry is 0.

**Non-example 2.16.** The matrix with rows (0 1 0), (0 0 1), (0 1 0) is not the adjacency matrix of a simple undirected graph, because $A_{12} = 1$ but $A_{21} = 0$. A matrix with a 1 on the diagonal (a loop) or an entry 2 (a double road) is not one either.

**In plain words.** *The table has some easy features. It looks the same when flipped over its diagonal. The diagonal holds only zeros. Adding up one row tells how many roads touch that house.*

**Proposition 2.17** (Properties of $A$)**.** *For the adjacency matrix $A$ of a simple undirected graph:*

1. *every entry is* 0 *or* 1*;*

2. *$A$ is symmetric: $A_{ij} = A_{ji}$;*

3. *the diagonal is zero: $A_{ii} = 0$;*

4. *the sum of row $i$, and of column $i$, equals $\deg(n_i)$;*

5. *the sum of all entries equals $2|E|$;*

6. *conversely, every $N \times N$ matrix of zeros and ones that is symmetric with zero diagonal is the adjacency matrix of some simple graph with numbered vertices.*

Items 1–3 and 6 are stated in the textbook [9, pp. 11–12], and items 2, 3 and 6 also appear in other notes [2, 7]. Items 4 and 5 are derived here; a published note states item 4 in a form that allows loops [7].

*Proof.* Items 1–3 follow from Definition 2.13, because adjacency is symmetric and no vertex is adjacent to itself. For item 4, the sum of row $i$ is the number of $j$ with $A_{ij} = 1$, which is the number of neighbours of $n_i$. Each neighbour is joined to $n_i$ by exactly one edge, since there are no repeated edges, and no edge at $n_i$ joins it to itself, since there are no loops. So the number of neighbours equals the number of edges at $n_i$, which is $\deg(n_i)$. Column $i$ gives the same number by symmetry. For item 5, add item 4 over all $i$. Each edge $n_i n_j$ adds one to $\deg(n_i)$ and one to $\deg(n_j)$, so it is counted twice. For item 6, join $n_i$ and $n_j$ exactly when the entry is 1. Symmetry makes this a set of unordered pairs, and the zero diagonal excludes loops. □

For the board matrix the row sums are $1, 2, 1$, the column sums are $1, 2, 1$, and the total is $4 = 2|E|$ with $|E| = 2$ (script, Appendix D).

## 2.3 Walks

**In plain words.** *A walk is a trip along the roads. Start at a house and move to a neighbour, again and again. You may pass the same house twice. You may drive the same road twice. The length of a trip is the number of roads you drove. If you drive a road twice, you count it twice. Staying at home is also a trip, and its length is zero.*

**Definition 2.18** (Walk; textbook Definition 1.2.10)**.** A *walk of length $k$ from $u_0$ to $u_k$*, also called a *$(u_0, u_k)$-walk*, is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_i = u_{i-1}u_i$ for $1 \le i \le k$ [9, p. 21]. The *length* of the walk is $k$, the number of edges, each edge counted as many times as it occurs [9, p. 22].

**Remark 2.19.** A walk allows repeated vertices and repeated edges [9, p. 22]. It forbids one thing only: two consecutive vertices that are not joined by an edge. A walk with $k$ edges lists $k + 1$ vertices. The textbook does not say in so many words that $k = 0$ is allowed. This paper reads $k = 0$ as allowed, because the textbook's property $d(u, u) = 0$ of distance needs a path of length 0 from $u$ to $u$ [9, p. 23], and another text has a length-0 walk as an example [3, Example 2.9.2]. This is an interpretation and is qualified.

**Remark 2.20** (A walk is its list of vertices)**.** In a simple graph there is at most one edge between two vertices, so the edge $e_i = u_{i-1}u_i$ is forced by its two ends. Hence a walk of length $k$ from $n_i$ to $n_j$ is the same thing as a sequence $w_0, w_1, \ldots, w_k$ of vertices with $w_0 = n_i$, $w_k = n_j$ and $w_{t-1}, w_t$ adjacent for each $t$ from 1 to $k$ (derived). Two walks are different exactly when these sequences differ at some position. From here on a walk is written by its vertices. For a graph with repeated edges this step would fail, which is one reason why "simple" matters.

**Example 2.21.** In the board graph the sequence $n_1, n_2, n_3, n_2$, written $1, 2, 3, 2$, is a walk of length 3 from $n_1$ to $n_2$. So is $1, 2, 1, 2, 3$, of length 4, from $n_1$ to $n_3$; it repeats the vertices 1 and 2 and the edge $n_1n_2$. The textbook's example is $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ in its graph of Fig. 1.2.5, a walk of length 8 [9, p. 22].

**Non-example 2.22.** $1, 3$ is not a walk of the board graph, because $n_1$ and $n_3$ are not adjacent. (It would be a walk of the drawing's triangle.) The textbook's non-example is $v_3, v_1, v_4, v_2, v_3, v_5$ [9, p. 22].

**The textbook graph.** For the textbook's examples the paper needs a graph with a cycle, and the board graph has none. The textbook's Fig. 1.2.5 shows a graph on five vertices $v_1, \ldots, v_5$. The textbook's own examples force the seven edges $v_1v_2$, $v_1v_3$, $v_2v_3$, $v_2v_4$, $v_3v_4$, $v_3v_5$, $v_4v_5$ [9, p. 22]. Only these seven edges are used below, and the graph they form is called the *textbook graph*; nothing depends on reading other lines of the figure. Its vertices are written $1, \ldots, 5$ in the sequences below.

**In plain words.** *A trip is closed when it ends at the house where it began. That is the only new rule. The trip may still pass houses and roads many times.*

**Definition 2.23** (Closed walk; textbook Definition 1.2.11)**.** A $(u_0, u_k)$-walk is *closed* if $u_0 = u_k$ [9, p. 22].

**Remark 2.24.** A closed walk allows everything that a walk allows and forbids only a different start and end. A walk of length 0 is closed. In a simple graph no closed walk has length 1, since that would be a loop, and a closed walk of length 2 has the form $u, w, u$ and uses one edge twice (derived).

**Example 2.25.** In the board graph $2, 1, 2$ is a closed walk of length 2, and so is $2, 3, 2, 1, 2$ of length 4.

**Non-example 2.26.** $1, 2, 3$ is not closed in the board graph: it starts at 1 and ends at 3.

**In plain words.** *Now add one rule. Never drive the same road twice. You may still pass the same house twice, if you come back by a different road. English books call such a trip a trail. Our textbook calls it a path. This paper says trail, so that the word path can keep one meaning.*

**Definition 2.27** (Trail; textbook Definition 1.2.12, which names it "path")**.** A $(u_0, u_k)$-walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct [9, p. 22].

The textbook calls this object a path and writes $|P|$ for its length [9, p. 22]; English usage calls it a trail [3, 8]. When the paper needs to point at the textbook's word it writes "textbook path".

**Remark 2.28.** A trail allows repeated vertices and forbids repeated edges. A vertex can recur in a trail only if the trip left it by one road and came back by another, so the trip has gone round a loop of the graph. In the board graph every trail is also a path in the sense of Definition 2.31 for every length from 0 to 4, and the board graph has no closed trail of positive length (script, Appendix D). So the board graph cannot show the difference between a trail and a path, and the textbook graph is used for that (derived, checked by script for lengths up to 4).

**Example 2.29.** In the textbook graph, $1, 3, 2, 4, 3, 5$ is a trail: its edges $v_1v_3, v_3v_2, v_2v_4, v_4v_3, v_3v_5$ are distinct, although the vertex $v_3$ occurs twice [9, p. 22].

**Non-example 2.30.** In the textbook graph, $1, 3, 2, 4, 3, 2, 4, 3, 5$ is a walk but not a trail, because the edge $v_3v_2$ occurs twice [9, p. 22]. In the board graph $1, 2, 1, 2, 3$ is not a trail, because the edge $n_1n_2$ occurs twice.

**In plain words.** *Now add a stronger rule. Never visit the same house twice. Then you also never drive the same road twice. A trip like this goes straight from start to end without any loop. Our course calls it a path.*

**Definition 2.31** (Path; textbook Definition 1.2.13, which names it "simple path")**.** A $(u_0, u_k)$-walk is a *path* if all its vertices $u_0, \dots, u_k$ are pairwise distinct [9, p. 22].

The textbook calls this a simple path and writes $P_n$ for the simple path on $n$ vertices [9, p. 22]. English texts call it a path [3, 8]. The textbook defines it as a trail with distinct vertices; Proposition 2.35 shows that distinct vertices already give distinct edges, so the two ways of saying it agree. That the *course* uses the word "path" in this sense comes from the assignment text; the board does not say which meaning it uses. So this reading is an interpretation, and it is qualified [9, p. 22].

**Remark 2.32.** A path forbids repeated vertices, and therefore repeated edges. It allows length 0 (a single vertex) and length 1 (a single edge). It does not allow a closed path of positive length, because the first and last vertices would be equal.

**Example 2.33.** In the board graph $1, 2, 3$ is a path of length 2 from $n_1$ to $n_3$. The textbook's example is $v_1, v_3, v_4, v_5$ in its graph [9, p. 22].

**Non-example 2.34.** In the board graph $1, 2, 3, 2$ is a walk but not a path, because the vertex 2 occurs twice. In the textbook graph $1, 3, 2, 4, 3, 5$ is a trail but not a path, because $v_3$ occurs twice [9, p. 22].

**In plain words.** *There is a quick reason why a clean route never uses a road twice. Look at any two roads on the route. The later road ends at a house that comes after both houses of the earlier road. That house is new, so the two roads differ.*

**Proposition 2.35.** *In a simple graph a walk with pairwise distinct vertices has pairwise distinct edges. Hence every path is a trail.*

This is stated as an exercise in one text [3, Exercise 2.9]; the proof here is derived.

*Proof.* Let $u_0, \dots, u_k$ have distinct vertices and take two steps with indices $a < b$, that is, the edges $u_{a-1}u_a$ and $u_{b-1}u_b$. The vertex $u_b$ has an index larger than $a$, so it differs from both $u_a$ and $u_{a-1}$. But $u_b$ is an end of the second edge and is not an end of the first. So the two edges are different. □

## 2.4 Cycles

**In plain words.** *A cycle is a closed trip that never drives the same road twice. Our textbook lets such a trip pass a house twice on the way. A simple cycle is stricter. It passes no house twice, except the start, which is also the end. A simple cycle is a ring of houses with a road between each pair of neighbours.*

**Definition 2.36** (Cycle; textbook Definition 1.2.14)**.** A *cycle* (the textbook also says *closed path*) is a closed trail: a $(u_0, u_k)$-walk with $u_0 = u_k$ and pairwise distinct edges [9, p. 22].

**Definition 2.37** (Simple cycle; textbook Definition 1.2.15)**.** A cycle is *simple* if only its first and last vertices repeat [9, p. 22]. The textbook writes $C_n$ for the simple cycle on $n \geq 3$ vertices [9, p. 22].

**Remark 2.38.** A cycle allows repeated vertices (a figure eight is a cycle, by the definition; derived) and forbids repeated edges and open walks. A simple cycle forbids any repeated vertex other than the closing one. In a simple graph a closed trail cannot have length 1 or 2, and this paper does not count a closed walk of length 0 as a cycle; the textbook is silent on that case (interpretation, qualified). English usage differs: one text defines a cycle as a closed walk with $k \geq 3$ and distinct vertices $w_0, \ldots, w_{k-1}$, which is the textbook's simple cycle [3, Definition 2.10.1(b)]. This paper follows the textbook for "cycle" and writes "simple cycle" where English texts write "cycle". Cycles are not needed for the main results; they are given because the textbook's list of definitions includes them.

**Example 2.39.** The drawing's triangle has the edges 12, 13, 23, and $1, 2, 3, 1$ is a simple cycle of length 3 in it. The textbook's example of a simple cycle is $v_1, v_3, v_4, v_2, v_1$ [9, p. 22]. The textbook's walk $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ is a cycle in the sense of Definition 2.36 (closed, six distinct edges) but not a simple one, since $v_3$ occurs three times [9, p. 22].

**Non-example 2.40.** In the board graph $1, 2, 1$ is a closed walk but not a cycle, because the edge $n_1n_2$ is used twice. In fact the board graph has no cycle at all: among its closed walks of lengths 1 to 6, none has distinct edges (script, Appendix D).

## 2.5 Three naming conventions

**In plain words.** *Different books use different names for the same kinds of trips. Here is one table that puts the names side by side. Our paper follows the last column.*

Table 3 lists the same properties under the course, textbook, and English names. The course column is the assignment's statement for "path" only; for "cycle" the course convention is not stated, so the paper follows the textbook.

Table 3: Naming conventions. "Textbook" gives the textbook's word and definition number [9]; "English" gives the usage of [3, 8].

| Property of the walk | Textbook | English | This paper |
|---|---|---|---|
| any steps along edges | walk (1.2.10) | walk | walk |
| first vertex equals last | closed walk (1.2.11) | closed walk | closed walk |
| no repeated edge; vertices may repeat | "path" (1.2.12) | trail | trail |
| no repeated vertex (so no repeated edge) | "simple path" (1.2.13) | path | **path** |
| closed, no repeated edge; vertices may repeat | "cycle" (1.2.14) | closed trail | cycle |
| closed, only first and last vertex equal, $k \geq 3$ | "simple cycle" (1.2.15) | cycle | simple cycle |

So the word "path" has three meanings across the sources: the textbook's Definition 1.2.12, the textbook's Definition 1.2.13 under the name "simple path", and the English usage. Another text records that some authors use "path" for repeated vertices and "simple path" for distinct vertices, which is the textbook's pattern [8]. So the course convention is one of at least two in use.

## 2.6 Distance

**In plain words.** *The distance between two houses is the number of roads on the shortest trip between them. If no trip exists, we say the distance is infinite. A house is at distance zero from*

*itself. The distance there equals the distance back, because any trip can be driven backwards. Going by a third house is never shorter than the shortest trip.*

**Definition 2.41** (Distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ between vertices $u$ and $v$ is the length of a shortest $(u, v)$-path if at least one exists, and $+\infty$ otherwise [9, pp. 22–23].

The textbook's "path" here is its Definition 1.2.12, a trail. The definition does not depend on whether one reads trail or path, because the shortest walk, the shortest trail, and the shortest path between two vertices have the same length (Proposition A.1, derived). The textbook lists three properties, for all vertices $u, v, w$ [9, p. 23].

1. $d(u, v) \geq 0$, and $d(u, v) = 0$ if and only if $u = v$;
2. $d(u, v) = d(v, u)$;
3. $d(u, v) \leq d(u, w) + d(w, v)$.

So when every two vertices are joined by a path, the pair $(V, d)$ is a metric space [9, p. 23]. The textbook states these properties without proof; they are derived from the definition in Appendix A.1.

**Remark 2.42.** The value $+\infty$ is not a whole number; it is how a missing path shows up. The definition forbids negative values and any value that is not the length of an actual trip.

**Example 2.43.** In the board graph $d(n_1, n_2) = 1$, $d(n_2, n_3) = 1$, $d(n_1, n_3) = 2$ and $d(n_i, n_i) = 0$ for each $i$ (script). The walk $1, 2, 1, 2, 3$ has length 4, but the distance is the shortest length, which is 2.

**Non-example 2.44.** 4 is not the distance from $n_1$ to $n_3$ in the board graph, even though a walk of length 4 exists. Also, in the graph with vertices $n_1, n_2, n_3$ and the single edge $n_1n_2$, the distance $d(n_1, n_3)$ is not a number: it is $+\infty$ (script).

## 2.7 Every walk contains a path

**In plain words.** *Suppose you can drive from one house to another. Then you can do it without visiting any house twice. Look at your trip. If it passes some house twice, you went round a loop. Cut the loop out. What is left is still a trip from the same start to the same end. It is shorter. Keep cutting until no house is repeated. The trip that remains is never longer than the first one.*

**Lemma 2.45** (Walk contains path; textbook Lemma 1.2.1 for $u \neq v$)**.** *Let $u$ and $v$ be vertices of a simple graph, equal or not. If there is a $(u, v)$-walk of length $k$, then there is a $(u, v)$-path of length at most $k$, and all its vertices and edges are vertices and edges of the walk.*

The textbook states the lemma only for $u \neq v$ and says that from any $(u, v)$-walk one can extract a simple $(u, v)$-path, without a bound on the length [9, p. 23]. Its proof is by induction on the length of the walk [9, p. 23], and the length bound follows from that proof. Another text states the version with $u = v$ allowed and with the bound [3, Corollary 2.9.7]. The case $u = v$ is needed here because a closed walk such as $2, 1, 2$ has $u = v$; the path it contains is the single vertex, of length 0. The proof below is the paper's own (derived).

*Proof.* We prove by strong induction on $k$ the statement $S(k)$: every $(u, v)$-walk of length $k$, for any vertices $u, v$, contains a $(u, v)$-path of length at most $k$.

*Base case,* $k = 0$. A walk of length 0 is a single vertex $u$, so $v = u$. A single vertex has no repeated vertex, so the walk is itself a path, of length $0 \leq 0$.

*Induction hypothesis.* Let $k \geq 1$ and assume $S(m)$ for every $m < k$.

*Step.* Let $u_0, u_1, \ldots, u_k$ be a $(u, v)$-walk, so $u_0 = u$ and $u_k = v$. If its vertices are pairwise distinct, it is a path of length $k \leq k$ and we are done. Otherwise some vertex occurs twice, so there are indices $i < j$ with $u_i = u_j$. Form the sequence $W' = u_0, \ldots, u_i, u_{j+1}, \ldots, u_k$; if $j = k$ nothing follows $u_i$. We check that $W'$ is a $(u, v)$-walk. (a) It starts at $u_0 = u$. (b) It ends at $v$: if $j < k$ its last vertex is $u_k = v$, and if $j = k$ its last vertex is $u_i = u_j = u_k = v$. (c) Consecutive vertices are adjacent: before the cut and after the join they are consecutive vertices of the old walk, and the join pair $(u_i, u_{j+1}) = (u_j, u_{j+1})$ is a step of the old walk. The length of $W'$ is $i + (k - j)$, which is less than $k$ because $i < j$. By the induction hypothesis $W'$ contains a $(u, v)$-path of length at most $|W'| < k$, hence at most $k$. Every vertex and edge of that path is one of $W'$, hence one of the old walk. □

*Why the proof is valid.* Each cut removes at least one step, so the length drops by at least one, and a walk that cannot be cut has no repeated vertex, which is the definition of a path. As a trace, the textbook's walk $1, 3, 2, 4, 3, 2, 4, 3, 5$ of length 8 has $u_1 = u_4 = 3$; cutting gives $1, 3, 2, 4, 3, 5$ of length 5; cutting again gives $1, 3, 5$ of length 2, which is a path (script). In the board graph $1, 2, 1, 2, 3$ cuts to $1, 2, 3$, and the closed walk $2, 1, 2$ cuts to the single vertex 2.

**In plain words.** *A clean route can have at most one road fewer than the number of houses. It visits each house at most once. So a clean route cannot be very long. If any trip exists, a short trip exists too.*

**Corollary 2.46** (Length bound)**.** *In a graph with $N$ vertices every path has at most $N - 1$ edges. Hence if some walk joins $u$ and $v$, then some walk of length at most $N - 1$ joins them.*

*Proof.* A path of length $k$ lists $k + 1$ pairwise distinct vertices, and there are only $N$ vertices, so $k + 1 \leq N$. For the second part, apply Lemma 2.45: the path inside the walk has length at most that of the walk and, by the first part, at most $N - 1$. □

The same argument is given in another text [3, Section 2.9.4]. In the board graph $N = 3$, so a path has at most 2 edges, and the longest path has 2 edges (script).

# 3 Q1: what the entry of a power of $A$ counts

## 3.1 Matrix products and powers

**In plain words.** *To multiply two tables of the same size, take a row of the first and a column of the second. Multiply them place by place, then add. That gives one number of the new table. To raise a table to a power, multiply it by itself again and again.*

**Definition 3.1** (Matrix product and power)**.** The *product* of two $N \times N$ matrices $X$ and $Y$ is the matrix $XY$ with $(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}$. The powers of $A$ are $A^1 = A$ and $A^k = A^{k-1} A$ for $k \geq 2$.

The exponent 0 is not used for $A$ in this section; by the usual convention $A^0 = I$, the identity matrix, which corresponds to one walk of length 0 from a vertex to itself [3]. It appears only in Section 4, in $(I + A)^{N-1}$.

**Remark 3.2.** The definition allows a sum over all $N$ vertices $r$, whether or not $r$ is a neighbour. It forbids multiplying entry by entry: the product is not the table of products of matching entries.

**Example 3.3.** For the board matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ the entry $(A^2)_{11} = A_{11}A_{11} + A_{12}A_{21} + A_{13}A_{31} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$ (script).

**Non-example 3.4.** Squaring every entry of $A$ separately gives back $A$, since the entries are 0 and 1. That is not $A^2$: the matrix product has $(A^2)_{22} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$ (script), where entry-wise squaring gives $A^2_{22} = 0$.

## 3.2 The walk-counting theorem

**In plain words.** *Take the table of roads and multiply it by itself. Each number in the new table counts trips of two roads. The trips run from the row house to the column house. Multiply again, and you count trips of three roads. The count includes trips that turn back. So the table counts trips, not clean routes.*

**Theorem 3.5** (Walk counting)**.** *Let $A$ be the adjacency matrix of a simple undirected graph with vertices $n_1, \ldots, n_N$ and let $k \geq 1$ be a whole number. Then $(A^k)_{ij}$ is the number of walks of length $k$ from $n_i$ to $n_j$, for all $i, j$.*

The theorem is a standard one. It is stated and proved for simple undirected graphs in lecture notes [2, Theorem 1.1.3], and for digraphs and multidigraphs, of which the simple undirected graph is the symmetric case, in two other texts [3, Theorem 4.5.10], [6, Corollary 9.3.3]; a fourth set of notes states it too [1, Theorem 1]. The course textbook does not state it in the pages checked, so it is proved here (qualified, as said in Section 1). The proof below follows the route that the standard notes use: induction on $k$, sorting the walks by their second-last vertex [2, 3].

**Remark 3.6.** The theorem allows $i = j$ (closed walks), and it counts every walk once, with repeats of vertices and edges permitted. It forbids reading the number as a count of paths (Proposition 3.11).

**Example 3.7.** For the board graph, $(A^2)_{13} = 1$, the walk $1, 2, 3$ (Section 6).

**Non-example 3.8.** $(A^2)_{12} = 0$ in the board graph. This does not say that $n_1$ and $n_2$ are unconnected. It says that no walk of length exactly 2 joins them; walks of length 1 and 3 do exist.

*Proof (derived; induction on $k$).* Write $W_k(i, j)$ for the number of walks of length $k$ from $n_i$ to $n_j$. We prove $W_k(i, j) = (A^k)_{ij}$ for all $i, j$, for each $k \geq 1$. By Remark 2.20 we count sequences of vertices.

*Base case, $k = 1$.* A walk of length 1 from $n_i$ to $n_j$ is a sequence $(n_i, n_j)$ with $n_i, n_j$ adjacent. There is exactly one such sequence if $n_i$ and $n_j$ are adjacent, and none otherwise. So $W_1(i, j) = A_{ij} = (A^1)_{ij}$.

*Induction hypothesis.* Let $k \geq 2$ and suppose $W_{k-1}(i, j) = (A^{k-1})_{ij}$ for all $i, j$.

*Inductive step.* Fix $i$ and $j$. Every walk $(w_0, \ldots, w_k)$ of length $k$ from $n_i$ to $n_j$ has a second-last vertex $w_{k-1}$. Because $k \geq 2$, deleting the last vertex leaves $w_0, \ldots, w_{k-1}$, a sequence with consecutive vertices adjacent and length $k - 1 \geq 1$, so the induction hypothesis speaks about walks of that length. Let $w_{k-1} = n_r$, where $r$ is one of $1, \ldots, N$. Sort the walks into $N$ classes by the value of $r$. The classes are disjoint, because each walk has exactly one second-last vertex, and together they contain every walk. So $W_k(i, j)$ is the sum over $r = 1, \ldots, N$ of the number of walks of length $k$ from $n_i$ to $n_j$ whose second-last vertex is $n_r$.

Fix $r$. We claim that the number in class $r$ is $W_{k-1}(i, r) \cdot A_{rj}$.

- If $n_r$ and $n_j$ are not adjacent ($A_{rj} = 0$), no walk can have $n_r$ followed by $n_j$, so class $r$ is empty and the count is $0 = W_{k-1}(i, r) \cdot 0$.

- If $n_r$ and $n_j$ are adjacent ($A_{rj} = 1$), delete the last vertex of a walk in class $r$. What remains is a sequence $w_0, \ldots, w_{k-1}$ with $w_0 = n_i$, $w_{k-1} = n_r$ and consecutive vertices adjacent, that is, a walk of length $k-1$ from $n_i$ to $n_r$. Conversely, any walk of length $k-1$ from $n_i$ to $n_r$, followed by $n_j$, is a sequence of $k+1$ vertices whose consecutive pairs are adjacent, because the new pair $n_r, n_j$ is adjacent. So it is a walk of length $k$ from $n_i$ to $n_j$ with second-last vertex $n_r$. Deleting and appending undo each other, so they match the walks of class $r$ one to one with the walks of length $k-1$ from $n_i$ to $n_r$, and the count is $W_{k-1}(i,r) = W_{k-1}(i,r) \cdot 1$.

Therefore $W_k(i,j) = \sum_{r=1}^{N} W_{k-1}(i,r)A_{rj}$. By the induction hypothesis $W_{k-1}(i,r) = (A^{k-1})_{ir}$, so $W_k(i,j) = \sum_r (A^{k-1})_{ir}A_{rj}$, which is the $(i,j)$ entry of $A^{k-1}A = A^k$ by Definition 3.1. This proves the statement for $k$, and by induction for all $k \geq 1$. □

*Why the argument is valid.* Two facts are the only places where the counting could go wrong: the sum over the second-last vertex covers each walk exactly once, and the matching in the case $A_{rj} = 1$ is one to one. Simplicity is used twice: to identify a walk with its vertex sequence (Remark 2.20), and so that $A_{rj}$, the number of edges between $n_r$ and $n_j$, is 0 or 1. Undirectedness is not used in the induction; it only gives the symmetry of $A$ (interpretation; the versions for digraphs in the cited texts agree [3, 6]). For a graph with repeated edges the same proof counts walks with multiplicity if $A_{rj}$ is the number of edges, but then the properties of Proposition 2.17 change (interpretation). A computer check supports the theorem but is not a proof: for all labelled simple graphs on $N = 2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs) a brute-force enumeration of walks agreed with the matrix entries in all 40, 360, 5120 and 76800 tested cases of $(k, i, j)$ (Appendix A.5).

### 3.3 Consequences

**In plain words.** *A count is never negative, and it is always a whole number. A positive count means that at least one such trip exists. Reverse a trip and you get a trip back. So the table of trip counts also looks the same when flipped over its diagonal.*

**Corollary 3.9.** *For a simple undirected graph and $k \geq 1$: (a) every entry of $A^k$ is a non-negative whole number, so a sum of powers of $A$ has a positive $(i,j)$ entry if and only if some summand does; (b) $(A^k)_{ij} > 0$ if and only if there is a walk of length exactly $k$ from $n_i$ to $n_j$; (c) $A^k$ is symmetric.*

*Proof.* Parts (a) and (b) hold because by Theorem 3.5 the entry is a count. For (c), $(A^k)^T = (A^T)^k = A^k$ because $A$ is symmetric. Alternatively, reversing the sequence of a walk matches the walks from $n_i$ to $n_j$ with the walks from $n_j$ to $n_i$. □

**In plain words.** *Take the trips of two roads that start and end at the same house. Each one goes out along a road and comes straight back. So there is one such trip for each neighbour. That is the number of roads at the house.*

**Proposition 3.10** (Diagonal of $A^2$)**.** $(A^2)_{ii} = \deg(n_i)$ *for every $i$. Hence the sum of the diagonal entries of $A^2$ is $2|E|$.*

The first statement is made in lecture notes [2]; the proof below and the second statement are derived here.

*Proof.* By Theorem 3.5, $(A^2)_{ii}$ counts the walks $n_i, n_r, n_i$ of length 2 from $n_i$ to $n_i$. Such a walk exists exactly when $n_r$ is a neighbour of $n_i$, so there is one for each neighbour, and the number of neighbours is $\deg(n_i)$ (Definition 2.9). Summing over $i$ and using Proposition 2.17, item 5, gives $2|E|$. □

For the board graph the diagonal of $A^2$ is $(1, 2, 1)$, the degrees, and the sum 4 equals $2|E|$ (script).

### 3.4 Walks are not paths

**In plain words.** *A trip may turn back and visit the same house twice. A clean route never visits a house twice. The table counts trips, so it counts turn-back trips too. A house with at least one neighbour always has a two-road trip that goes out and comes back. So the table does not count clean routes.*

**Proposition 3.11** (Walks are not paths)**.** *If $n_i$ has at least one neighbour, then $(A^2)_{ii} = \deg(n_i) > 0$, but no path, and no trail, of length 2 from $n_i$ to $n_i$ exists. Hence $(A^k)_{ij}$ is in general not the number of paths, or of trails, of length $k$ from $n_i$ to $n_j$.*

*Proof.* The first inequality is Proposition 3.10. A path from $n_i$ to $n_i$ has its first and last vertices equal, so it can only be the single vertex, of length 0. A closed walk of length 2 has the form $n_i, n_r, n_i$ and uses the edge $n_i n_r$ twice, so it is not a trail. □

The warning holds under both of the textbook's readings of "path": distinct vertices (Definition 1.2.13) and distinct edges (Definition 1.2.12) [9, p. 22]. The general fact that the theorem counts walks and not paths is also made in the notes cited above [1, 3].

For the board graph this is concrete: $(A^2)_{11} = 1 > 0$, and the one walk is $1, 2, 1$, which repeats the vertex $n_1$ and the edge $n_1 n_2$. Also $(A^3)_{12} = 2$, the walks $1, 2, 1, 2$ and $1, 2, 3, 2$, and neither is a path or a trail, yet the only path from $n_1$ to $n_2$ has length 1. The enumeration for $k = 1, \ldots, 5$ finds $1, 0, 2, 0, 4$ walks from $n_1$ to $n_2$, of which $1, 0, 0, 0, 0$ are paths (script, Appendix A.2). So the entries grow with $k$ while the number of paths does not.

What a positive entry does guarantee is a walk of length $k$, and Lemma 2.45 then gives a path of length at most $k$ (derived). This is the fact that Section 4 uses.

### 3.5 The answer to Q1

For a simple undirected graph with adjacency matrix $A$ and any whole number $k \geq 1$, the entry in row $i$ and column $j$ of $A^k$ is the number of walks of length $k$ from $n_i$ to $n_j$. Walks may repeat vertices and edges, so the entry is not a count of paths.

## 4 Q2: deciding connectivity from the matrix

### 4.1 Connected graphs

**In plain words.** *Look at a village. Sometimes you can reach every house from every other house. You may pass the same house twice on the way. Then the village is called connected. If one house is cut off, the village is not connected.*

**Definition 4.1** (Connected graph; textbook Definition 2.1.1)**.** A graph $G = (V, E)$ is *connected* if for any two vertices $u$ and $v$ of $G$ there is a $(u, v)$-path in $G$ [9, p. 48].

The board states the same definition: a graph is called connected if for any $n_i, n_j$ there exists a path between the nodes [4, 5].

**Remark 4.2.** 1. In the textbook the word "path" here is Definition 1.2.12, a trail. The board does not say which meaning it uses, and the course convention is "no repeated vertex". The definition does not depend on the choice: a walk between two different vertices contains a path with distinct vertices (Lemma 2.45), and a path has distinct edges (Proposition 2.35), so both readings accept exactly the same graphs (derived; compare the cited lemma in other notes [3]).

2. "Any two vertices" includes $u = v$. For $u = v$ the walk with one vertex and no edges is a $(u, u)$-path, so these pairs are always satisfied, and reading the definition as "any two different vertices" changes nothing (interpretation; the textbook does not discuss $u = v$).

3. The definition forbids exactly one thing: a pair of vertices with no trip between them.

**Example 4.3.** The board graph is connected. The pair $n_1, n_3$ is joined by the path $1, 2, 3$, and the pairs $n_1, n_2$ and $n_2, n_3$ are joined by one edge. The textbook's own examples of connected graphs are the complete graphs, the complete bipartite graphs, and the three graphs of its Fig. 2.1.1 [9, p. 48].

**Non-example 4.4.** Three vertices $n_1, n_2, n_3$ with only the edge $n_1n_2$. No walk leaves $n_3$, so the pair $n_1, n_3$ has no path. The textbook's Fig. 2.1.2 shows a graph that is not connected and has four components [9, p. 49].

## 4.2 Components

**In plain words.** *Look at a village that is not connected. It falls into pieces. Inside one piece every house can reach every other house. No road joins two different pieces. Each piece is called a component. A connected village has exactly one piece.*

**Definition 4.5** (Connected component; textbook, printed p. 49, no number)**.** On the vertex set $V$ write $u \sim v$ if $G$ has a $(u, v)$-path. This relation is reflexive, symmetric and transitive, so $V$ splits into classes $V_1, \ldots, V_p$. The subgraphs $G_j = G[V_j]$ that these classes induce are the *connected components* of $G$ [9, p. 49].

The textbook gives this in a paragraph after Definition 2.1.1 without a number; the notation $G[V_j]$ means the graph on $V_j$ with all the edges of $G$ that join two vertices of $V_j$. Another text defines a connected graph as one with exactly one component, and shows that being joined by a walk and being joined by a path are the same relation [3, Definition 2.9.12, Proposition 2.9.10].

**Remark 4.6.** A component cannot be enlarged: a vertex reachable from a vertex of $V_j$ lies in $V_j$. The textbook's Remark 2.1.1 says that $G$ is connected if and only if it has exactly one component [9, p. 49].

**Example 4.7.** The graph with only the edge $n_1n_2$ has the two components $G[\{n_1, n_2\}]$ and $G[\{n_3\}]$. The board graph has the single component $G[\{n_1, n_2, n_3\}]$ (derived; Section 6).

**Non-example 4.8.** $G[\{n_1\}]$ is not a component of the graph with the edge $n_1n_2$ only, although it is a connected graph: $n_2$ can be reached from $n_1$, so it belongs to the same class. A component is a whole class, not any connected piece.

## 4.3 The sum used by the test

**In plain words.** *We need a name for a running total. It adds the counts of trips with one road, with two roads, and so on. We stop one step short of the number of houses. The whole check uses this one total.*

**Definition 4.9** (The sum $S$; new notation, not from the textbook)**.** For a graph with $N \geq 2$ vertices let $S = A + A^2 + \cdots + A^{N-1}$, the sum of the first $N - 1$ powers of $A$, added entry by entry.

**Remark 4.10.** By Theorem 3.5, $S_{ij}$ is the number of walks from $n_i$ to $n_j$ whose length is between 1 and $N - 1$. Hence every entry of $S$ is a non-negative whole number (derived). The diagonal of $S$ counts closed walks and plays no role in the test (Remark 4.21).

**Example 4.11.** For the board graph $S = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ (script; Section 6).

**Non-example 4.12.** $A^2$ alone is not $S$ for $N = 3$, where $S = A + A^2$. Either one alone gives a wrong answer for the board graph: $A^2$ has the zero entry $(1, 2)$ although $n_1$ and $n_2$ are joined by an edge, and $A$ has the zero entry $(1, 3)$ although $n_1$ and $n_3$ are connected (script).

## 4.4 The test and its proof

**In plain words.** *We want a quick check for a village. We only have the table that says which houses share a road. From the table we can count trips. Count the trips with one road, with two roads, and so on. Add the counts for each pair of houses. If no pair of different houses has a total of zero, the village is connected. We never need trips longer than one road fewer than the number of houses. A trip that wastes time can always be shortened. So longer trips tell us nothing new. A second form lets every house also stay put for one step. It gives the same answer.*

**Proposition 4.13** (One pair at a time)**.** *Let $G$ have $N \geq 2$ vertices and let $i \neq j$. The following three statements are equivalent.*

*(i) There is an $(n_i, n_j)$-path in $G$ (in the sense of textbook Definition 2.1.1).*

*(ii) There is a whole number $k$ with $1 \leq k \leq N - 1$ and $(A^k)_{ij} > 0$.*

*(iii) $S_{ij} > 0$.*

**Proposition 4.14** (The $I + A$ form)**.** *Let $G$ have $N \geq 2$ vertices, let $m \geq N - 1$ be a whole number, and let $i \neq j$. Then $\left((I + A)^m\right)_{ij} > 0$ if and only if $S_{ij} > 0$. Every diagonal entry of $(I + A)^m$ is at least 1.*

**Theorem 4.15** (The connectivity test)**.** *Let $G$ be a graph with $N \geq 2$ vertices and adjacency matrix $A$. The following are equivalent.*

*(a) $G$ is connected.*

*(b) Every off-diagonal entry of $S = A + A^2 + \cdots + A^{N-1}$ is positive.*

*(c) Every entry of $(I + A)^{N-1}$ is positive.*

The sum form is stated in a published note and in an encyclopedia entry [7, 11], and the second form is stated in lecture notes [1]. The encyclopedia entry gives no proof [11]. The note applies a walk-counting statement that says "paths" where walks are meant, and passes from a positive entry to a path without the walk-to-path lemma [7]. The lecture notes have only a sketch that again says "paths" [1]. The statements of the test are correct. So the proofs here are the paper's own (derived), built on Theorem 3.5 and Lemma 2.45. The chain of premises is listed in Appendix A.3.

**Remark 4.16.** The test allows any graph with at least two vertices, connected or not. It forbids using only one power, or stopping before the power $N - 1$ (Proposition 4.23), or reading the diagonal of $S$ (Remark 4.21).

**Example 4.17.** For the board graph the matrix $S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ has no zero entry, so by Theorem 4.15(b) the graph is connected (Section 6).

**Non-example 4.18.** For the graph with only the edge $n_1n_2$, $S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ has zeros off the diagonal, so the graph is not connected (Section 6).

*Proof of Proposition 4.13. (i) implies (ii).* A path is a walk, so there is a walk from $n_i$ to $n_j$. By Lemma 2.45 there is a path $P$ from $n_i$ to $n_j$ whose length $k$ is at most the length of that walk. $P$ has $k$ edges and therefore $k+1$ distinct vertices, and the graph has only $N$ vertices, so $k \leq N-1$ (Corollary 2.46). If $k$ were 0, $P$ would start and end at the same vertex, but $n_i \neq n_j$, so $k \geq 1$. $P$ is a walk of length $k$ from $n_i$ to $n_j$, so by Theorem 3.5 the number $(A^k)_{ij}$ is at least 1.

*(ii) implies (iii).* $S_{ij} = A_{ij} + (A^2)_{ij} + \cdots + (A^{N-1})_{ij}$. Each term is a count, hence a non-negative whole number (Corollary 3.9). A sum of non-negative numbers is at least as large as any one of its terms, and the term with the index $k$ from (ii) is positive, so $S_{ij} > 0$.

*(iii) implies (ii).* If $S_{ij} > 0$, a sum of non-negative terms is positive, so at least one term $(A^k)_{ij}$ with $1 \leq k \leq N-1$ is positive.

*(ii) implies (i).* If $(A^k)_{ij} > 0$ with $k \geq 1$, then by Theorem 3.5 there is at least one walk of length $k$ from $n_i$ to $n_j$. As $n_i \neq n_j$, Lemma 2.45 gives a path from $n_i$ to $n_j$ with distinct vertices. By Proposition 2.35 it also has distinct edges, so it is a $(n_i, n_j)$-path in either reading of the word "path" (Remark 4.2). □

*Proof of Proposition 4.14.* The proof expands $(I + A)^m$ with the binomial formula and is given in Appendix A.4. □

*Proof of Theorem 4.15.* Statement (a) says that for every pair of vertices there is a path between them. For a pair with $u = v$ this is always true (Remark 4.2). For pairs with $i \neq j$ it is statement (i) of Proposition 4.13. So (a) holds if and only if (i) holds for every $i \neq j$. By Proposition 4.13 this is equivalent to (iii) for every $i \neq j$, which is (b). By Proposition 4.14 with $m = N-1$ it is also equivalent to the positivity of every off-diagonal entry of $(I + A)^{N-1}$. Together with the diagonal entries, which are always positive by Proposition 4.14, this is (c). □

**Corollary 4.19** (Components from $S$)**.** *For any graph, connected or not, with $N \geq 2$ vertices and $i \neq j$: $S_{ij} > 0$ if and only if $n_i$ and $n_j$ lie in the same component.*

*Proof.* Proposition 4.13 uses connectivity only through the pair $(n_i, n_j)$: statement (i) says that $n_i \sim n_j$, which is the definition of "same class" in Definition 4.5. □

**Remark 4.20** (Distance)**.** A shortest walk between two different vertices is a path, by Lemma 2.45 and Proposition A.1. So the smallest $k$ with $(A^k)_{ij} > 0$ is the distance $d(n_i, n_j)$ of Definition 2.41 (derived); this is also stated in two other sources [6, 7].

**Remark 4.21** (The diagonal, and $N = 1$, $N = 2$)**.** For $N = 2$ and one edge, $S = A$ has zeros on the diagonal although the graph is connected (script). So the diagonal of $S$ cannot be required to be positive. For $N \geq 3$ and a connected graph the diagonal of $S$ is positive anyway, because every vertex has a neighbour and $(A^2)_{ii} = \deg(n_i)$ (Proposition 3.10), but the test does not need this. For $N = 1$ there are no pairs of different vertices, the graph is connected, and both forms of the test give the right answer, but $S$ is an empty sum, so the test is stated for $N \geq 2$ (derived).

## 4.5 The test as a procedure

**In plain words.** *Here is the check, step by step. First write the table of the village. Then multiply the table by itself again and again. After each step, add the new counts to a running total. Stop after one step fewer than the number of houses. Look at every pair of different houses.*

*If any total is zero, those two houses cannot reach each other. If no total is zero, everyone can reach everyone.*

**Test 4.22** (Matrix test)**.** Input: *the adjacency matrix $A$ of a graph with $N \geq 2$ vertices.*

1. *Set $P = A$ and $S = A$.*
2. *For $k = 2, 3, \ldots, N-1$: replace $P$ by the product $PA$ (now $P = A^k$) and add $P$ to $S$.*
3. *Look at the entries $S_{ij}$ with $i \neq j$. If one of them is $0$, the graph is not connected. If none is $0$, the graph is connected.*

*An equivalent form computes $(I + A)^{N-1}$ and reports "connected" if no entry is $0$.*

Correctness is Theorem 4.15. A zero at $(i, j)$ also says, by Corollary 4.19, that $n_i$ and $n_j$ lie in different components.

## 4.6 Why $N - 1$ powers are enough and $N - 2$ are not

Enough: Proposition 4.13 shows that if a trip between two different vertices exists at all, a trip of length at most $N - 1$ exists, because a shortest trip repeats no vertex and a repeat-free trip has at most $N$ vertices. So adding powers beyond $N-1$ cannot change which off-diagonal entries are positive. Not fewer: the next statement shows that $N - 1$ cannot be lowered.

**In plain words.** *Take four houses in a row. To go from the first house to the last, you must pass every house in between. That takes three roads. If we counted only trips of up to two roads, we would miss this pair. We would say the village is cut in two, and we would be wrong. The same idea works for any row of houses.*

**Proposition 4.23** ($N - 1$ cannot be lowered to $N - 2$)**.** *For every $N \geq 3$ there is a connected graph with $N$ vertices for which the sum $A + \cdots + A^{N-2}$ has a zero off-diagonal entry.*

*Proof.* Take the graph with edges $n_1n_2, n_2n_3, \ldots, n_{N-1}n_N$. Each step of a walk changes the vertex number by exactly 1. To go from $n_1$ to $n_N$ the number must increase by $N - 1$, so at least $N - 1$ steps are needed. So no walk of length $k < N - 1$ from $n_1$ to $n_N$ exists, and by Theorem 3.5 $(A^k)_{1N} = 0$ for $1 \leq k \leq N - 2$. The $(1, N)$ entry of $A + \cdots + A^{N-2}$ is therefore 0, although the graph is connected. □

A script confirms it for $N = 3, \ldots, 8$: the first power with a nonzero $(1, N)$ entry is $A^{N-1}$ each time (Appendix A.5). For $N = 3$ the graph is the board graph and the sum is $A$ itself, with $A_{13} = 0$. For $N = 4$ the matrices are written out in Section 6.

## 4.7 Breadth-first search, for comparison

**In plain words.** *There is another way, and it needs no table of counts. Stand at one house and mark it. Then mark every house next to it. Then mark every house next to those. Keep going until no new house appears. If every house is marked, the village is connected.*

**Remark 4.24** (Breadth-first search)**.** *Breadth-first search* (BFS) from a start vertex $s$ keeps a queue of discovered vertices, and repeatedly takes the vertex at the front and discovers its undiscovered neighbours. Its correctness statement in the source is that for every vertex $v$ there is a walk from $s$ to $v$ if and only if $v$ is discovered when the search ends [10]. So $G$ is connected if and only if every vertex is discovered when BFS starts from any single vertex $s$ (Remark 4.2 lets walks stand for paths). BFS also yields the components directly. The matrix test and BFS agree on all 32768 labelled graphs with 6 vertices and on all smaller ones (script, Section 6); this is a check, not a proof. The cost of the two methods is compared in Section 7.4.

### 4.8 The answer to Q2

A graph with $N \geq 2$ vertices and adjacency matrix $A$ is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 4.15).

## 5 Q3: the definition of a path between $n_i$ and $n_j$

**In plain words.** *Pick two houses of the village. A path between them is a clean route. It starts at the first house and ends at the second. It never visits any house twice. If the two houses are the same house, the route is just staying home.*

**Definition 5.1** (Path between $n_i$ and $n_j$; course convention)**.** Let $n_i$ and $n_j$ be vertices of $G$. A *path between $n_i$ and $n_j$* is a walk $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$ and $w_k = n_j$ in which the vertices $w_0, \ldots, w_k$ are pairwise distinct. Its length is $k$. For $i = j$ the only such path is the single vertex $n_i$, of length 0.

This is a simple path in the sense of textbook Definition 1.2.13 and a path in English usage [9, p. 22]. It is the specialization of Definition 2.31 to the endpoints $n_i$ and $n_j$. The statement that the course uses "path" in this sense comes from the assignment text; the board does not say which meaning it uses [4, 5]. The definition is therefore an interpretation, and it is qualified.

**Remark 5.2.** The definition allows length 0 (when $i = j$), length 1 (an edge), and at most $N - 1$ edges (Corollary 2.46). It forbids any repeated vertex, and therefore any repeated edge (Proposition 2.35). It does not forbid several different paths between the same two vertices; it says only what a path is.

**Example 5.3.** In the board graph there is exactly one path between $n_1$ and $n_3$, namely $1, 2, 3$. Every path has at most $N - 1 = 2$ edges, so the enumeration of walks up to length 4 finds all of them (script). Between $n_1$ and $n_1$ the only path is the one of length 0.

**Non-example 5.4.** In the board graph $1, 2, 1, 2, 3$ is a walk between $n_1$ and $n_3$ but not a path between them: it repeats the vertices 1 and 2. It is not a trail either, because it repeats the edge $n_1n_2$.

**The two textbook readings give the same answers.** If "path" is read as the textbook's Definition 1.2.12 (a trail), then "there is a path between $n_i$ and $n_j$" has the same truth value as in Definition 5.1. A path is a trail (Proposition 2.35), so a path gives a trail. A trail is a walk, and a walk contains a path (Lemma 2.45), so a trail gives a path. Hence every sentence of the form "there is a path between $n_i$ and $n_j$", in particular the definition of connected, has the same meaning in both readings (derived). What changes with the reading is a *count*: the number of trails between two vertices is not the number of paths, and neither is the number of walks (Proposition 3.11).

**The answer to Q3.** A path between $n_i$ and $n_j$ is a walk from $n_i$ to $n_j$ in which all vertices are pairwise distinct.

## 6 Worked examples

Every matrix, sum, and count in this section was computed by script (Appendix D); the matrices are written out entry by entry so that they can be checked against the graphs.

## 6.1 The board graph and its powers

**In plain words.** *We take the village from the board. It has three houses in a row. The first and last houses each have one road. The middle house has two roads. We count trips of two roads, then of three roads. We list every trip, so the reader can see that each count is right.*

The graph has the vertices $n_1, n_2, n_3$ and the edges $n_1n_2$ and $n_2n_3$; this is the graph of the board matrix (Section 1.4). Its adjacency matrix and the powers up to $A^5$ are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

$$A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \quad A^5 = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

Also $A^3 = 2A$ and $A^{k+2} = 2A^k$ for $k = 1, \ldots, 5$ (script). The matrix $A$ is symmetric with zero diagonal, and the row sums $1, 2, 1$ are the degrees (Proposition 2.17).

*Explicit walks* (derived by enumeration, labels $1, 2, 3$ for $n_1, n_2, n_3$).

- Length 1: $1, 2$ and $2, 3$ and their reverses.
- Length 2: from 1 to 1: $1, 2, 1$ (one walk). From 1 to 3: $1, 2, 3$ (one). From 2 to 2: $2, 1, 2$ and $2, 3, 2$ (two). From 3 to 3: $3, 2, 3$ (one).
- Length 3: from 1 to 2: $1, 2, 1, 2$ and $1, 2, 3, 2$ (two). From 2 to 3: $2, 1, 2, 3$ and $2, 3, 2, 3$ (two).

Each count equals the corresponding matrix entry, as Theorem 3.5 says; the script asserts this for every entry of $A$, $A^2$ and $A^3$.

*The sums from the inductive step.* The step of the proof of Theorem 3.5 adds up one term for each choice of the second-last vertex $n_r$. For $(A^3)_{12}$ the terms are $(A^2)_{1r}A_{r2}$ for $r = 1, 2, 3$, namely $1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$. The term for $r = 1$ is the walk $1, 2, 1$ followed by the step to $n_2$, giving $1, 2, 1, 2$; the term for $r = 3$ is the walk $1, 2, 3$ followed by the step to $n_2$, giving $1, 2, 3, 2$. Likewise $(A^2)_{22} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$ and $(A^2)_{11} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$.

*Counts by walk length.* Table 4 counts, over all start and end vertices, the walks, trails, paths and closed walks of each length $k$ by enumeration. The entries of $A^k$ for $k = 1, \ldots, 5$ agree with the enumeration (script).

Table 4: Walks, trails, paths and closed walks of length $k$ in the board graph, over all start and end vertices (script `m2_walks.py`).

| $k$ | walks | trails | paths | closed walks |
|---|---|---|---|---|
| 0 | 3 | 3 | 3 | 3 |
| 1 | 4 | 4 | 4 | 0 |
| 2 | 6 | 2 | 2 | 4 |
| 3 | 8 | 0 | 0 | 0 |
| 4 | 12 | 0 | 0 | 8 |

From $n_1$ to $n_3$ there are three walks of length at most 4: $1, 2, 3$ (length 2), and $1, 2, 1, 2, 3$ and $1, 2, 3, 2, 3$ (length 4); only the first is a path (script). The numbers of walks from $n_1$ to $n_2$ for $k = 1, \ldots, 5$ are $1, 0, 2, 0, 4$, and the numbers of paths among them are $1, 0, 0, 0, 0$ (script), as used in Proposition 3.11.

*The drawing.* The drawing on the board also has the line 1–3 and an arrowhead on the line 1–2. If the line 1–3 were an edge, the triangle $1, 2, 3, 1$ would be a simple cycle (Example 2.39), and the matrix would have $A_{13} = A_{31} = 1$. It does not, and this paper follows the matrix.

### 6.2 A connected example

**In plain words.** *Take three houses in a row. A road joins the first and the second house. Another road joins the second and the third house. The first and the third house share no road. Can you still go from one to the other? Yes, through the middle house. We expect the check to say connected.*

**Example 6.1** (Connected; the board graph, $N = 3$, $N - 1 = 2$)**.**

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad S = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

$S$ has no zero entry off the diagonal (it has none anywhere), so by Theorem 4.15(b) the graph is connected. $(I + A)^2$ has no zero entry, so by Theorem 4.15(c) the graph is connected. The two answers agree. The entry $A_{13}$ is 0, so no road joins $n_1$ and $n_3$, yet $(A^2)_{13} = 1$: there is exactly one walk of length 2 from $n_1$ to $n_3$, namely $1, 2, 3$. The $(1, 3)$ entry of $(I + A)^2$ is $1 = \binom{2}{2}(A^2)_{13}$, since the other two terms vanish there, and $(I + A)^2 = I + 2A + A^2$ holds entry by entry (script). BFS from vertex 1 reaches $\{1, 2, 3\}$ (script).

### 6.3 A disconnected example

**In plain words.** *Now take three houses again. Only one road exists, between the first and second house. The third house stands alone. Nobody can walk to it. We expect the check to say not connected. The check should also show that the third house is a piece by itself.*

**Example 6.2** (Disconnected; edge $n_1n_2$ only, $N = 3$, $N - 1 = 2$)**.**

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The off-diagonal zeros of $S$ are at $(1, 3)$, $(2, 3)$, $(3, 1)$ and $(3, 2)$, and the zeros of $(I + A)^2$ are at the same four places. So by Theorem 4.15 the graph is not connected, and by Corollary 4.19 the vertices $n_1$ and $n_2$ lie in one component and $n_3$ in another: the components are $\{n_1, n_2\}$ and $\{n_3\}$, which agrees with BFS (script). Higher powers do not help: the entries $(1, 3)$ and $(2, 3)$ of $A^k$ are 0 for $k = 1, \ldots, 6$, and in fact no walk reaches $n_3$ at all.

### 6.4 The bound $N - 1$ is needed

**In plain words.** *Now take four houses in a row. The first and the last house are three roads apart. So a check that stops at trips of two roads cannot see that they are connected. We write out the tables and watch what happens.*

**Example 6.3** (The path on four vertices, $N = 4$)**.** The edges are $n_1n_2$, $n_2n_3$, $n_3n_4$.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 2 & 0 \end{pmatrix}.$$

$$A+A^2=\begin{pmatrix}1&1&1&0\\1&2&1&1\\1&1&2&1\\0&1&1&1\end{pmatrix}\quad(N-2\text{ powers}),\qquad A+A^2+A^3=\begin{pmatrix}1&3&1&1\\3&2&4&1\\1&4&2&3\\1&1&3&1\end{pmatrix}\quad(N-1\text{ powers}).$$

With $N-2=2$ powers the $(1,4)$ entry is 0, and the test would wrongly say "not connected". With $N-1=3$ powers every entry is positive. This is the case $N=4$ of Proposition 4.23.

### 6.5 Exhaustive agreement with breadth-first search

**In plain words.** *Examples on a few villages are not proof. So we also ran the check on every possible village with up to six houses. Each time we compared it with the marking method. The two methods always gave the same answer.*

Table 5 reports a script that built every labelled simple graph with $N=2,\dots,6$ vertices, ran the sum test and the $(I+A)^{N-1}$ test, and compared them with an independent BFS. The last column shows what goes wrong with only $N-2$ powers. This is a test of the statements and not a replacement for the proofs in Section 4, and it covers $N\le 6$ only.

Table 5: Exhaustive check over all labelled simple graphs with $N$ vertices (script `m3_connectivity.py`). Columns "sum test" and "$(I+A)^{N-1}$ test" give the number of graphs on which the test agrees with BFS.

| $N$ | graphs | connected | sum test agrees | $(I+A)^{N-1}$ test agrees | wrong with $N-2$ powers |
|---|---|---|---|---|---|
| 2 | 2 | 1 | 2 | 2 | 1 |
| 3 | 8 | 4 | 8 | 8 | 3 |
| 4 | 64 | 38 | 64 | 64 | 12 |
| 5 | 1024 | 728 | 1024 | 1024 | 60 |
| 6 | 32768 | 26704 | 32768 | 32768 | 360 |

For $N=3$ the $N-2$ test is wrong on 3 of the 8 graphs, the three labelled paths, and for $N=4$ on 12 of 64. For $N=2$ the sum with $N-2=0$ powers is empty and is wrong on the one connected graph. The same script also confirmed that the statement of Corollary 4.19 and the repeated-squaring version of the test (Section 7.4) agree with BFS on all these graphs.

## 7 Discussion: limits, disagreements, and cost

### 7.1 What is proved and what is only checked

The two main results, Theorem 3.5 and Theorem 4.15, are proved in full from the definitions, using Lemma 2.45 for the second. The computer checks are tests of the statements and not proofs: the walk count was compared with brute-force enumeration for all labelled simple graphs on 2 to 5 vertices, and the test was compared with BFS for all labelled simple graphs on 2 to 6 vertices (Appendix A.5). The proof of Theorem 4.15 is only as good as Theorem 3.5 and Lemma 2.45. The first is stated by four sources [2, 3, 6, 1]. The second combines the textbook's lemma, which needs $u\neq v$ and states no length bound, with a version from another text that allows $u=v$ and gives the bound [9, 3]; the paper proves the combined form itself. The arithmetic is ordinary integer arithmetic. Entries are counts and hence non-negative, which is what makes "a sum is positive if and only if some term is positive" valid; with subtraction or modular arithmetic that step would fail (derived).

### 7.2 Conventions that change the words but not the mathematics

- *Path.* Under the textbook's own word, "path" allows repeated vertices (Definition 1.2.12), and a reader who uses it will count trails [9]. The paper states which meaning it uses and keeps it (Table 3). The board does not say which meaning it uses, and the course reading comes from the assignment text. This does not change the connectivity result (Remark 4.2), and it changes only counts (Proposition 3.11).

- *Cycle.* The textbook's cycle is a closed trail and the English cycle is the textbook's simple cycle [9, 3]. Cycles are not needed for the main results.

- *Length* 0. Whether a walk of length 0 is allowed, and whether a closed trail of length 0 is a cycle, is not stated in the textbook pages read (Remark 2.19, Remark 2.38). For $i \neq j$ the connectivity result needs only paths of length at least 1.

- *The mark above A.* If the mark is an $n$ and not a $k$, Theorem 3.5 still describes $(A^n)_{ij}$, so the mathematics is unchanged; the reading of the board remains open [4, 5].

- *The drawing.* The drawing and the matrix on the board disagree on the line 1–3 and on an arrowhead (Section 1.4). The paper follows the matrix and does not guess which one the lecturer intended.

### 7.3 Where the sources disagree

One published note states its Theorem 1 as "the entry counts different paths of length $l$", while its own property 7 speaks of walks [7]. As written, Theorem 1 is false for any definition of path that forbids repeated vertices or repeated edges, by Proposition 3.11. This paper does not cite that theorem as support for Theorem 3.5; it cites the notes that say walks [2, 3, 6, 1]. The note and the lecture notes that state the connectivity test explain it by "paths" where walks are meant, so their explanation is wrong although their statements of the test are correct (Section 4) [7, 1]. Eigenvalue-based tests are outside the scope of this note, and nothing is claimed about them.

### 7.4 Cost of the matrix test and of breadth-first search

This subsection is an estimate that counts operations and does not time programs. Take the school method, in which one product of two $N \times N$ matrices takes $N^3$ scalar multiplications. Reaching $A^{N-1}$ from $A$ takes $N-2$ products, so the matrix test costs about $(N-2)N^3$ multiplications: 98,000,000 for $N = 100$ (script). By Proposition 4.14 the exponent $N-1$ may be replaced by any larger exponent, so one can square $I + A$ repeatedly until the exponent $2^t$ is at least $N-1$. That takes $t$ squarings, 7 for $N = 100$, costing 7,000,000 multiplications (script); the script checked this version against BFS on every labelled graph with at most 6 vertices. BFS takes time proportional to $n + m$, the numbers of vertices and edges, when the graph is given by adjacency lists [10]. When the input is the matrix $A$ itself, BFS reads each row once, which is $N^2 = 10{,}000$ entries for $N = 100$ (derived). The entries of the powers also grow fast: for the complete graph on 10 vertices the largest entry of $A^9$ is 38,742,049 (script). The matrix test is useful because its answer is a clean statement about powers of $A$, not because it is fast. Faster matrix multiplication methods were not examined. The cited slides compare BFS with no matrix test, and the count of $N-2$ products is this paper's own [10].

### 7.5 Limits and open questions

- The theorems are for simple graphs. With loops or repeated edges the entry $A_{rj}$ may exceed 1 and $A_{ii}$ may be nonzero; the same proof then counts walks with multiplicity

(interpretation), but Proposition 2.17 changes. The test is not claimed for directed graphs: no source was checked for them, and the symmetry of $A$ is used.

- The textbook states no walk-counting theorem and no matrix connectivity test in the pages read, and a search of the book found none; the rest of the book was not read page by page, so this absence is qualified [9].
- The brute-force checks cover small graphs only.
- The test says connected or not. The sum $S$ does not list a path; Corollary 4.19 gives the components and Remark 4.20 the distances.
- Open: whether the lecturer's intended answer to the board's question 2 was the sum test, the $(I + A)^{N-1}$ test, or BFS; the board shows only the question.
- Open: the reading of the mark above $A$, and the mismatch between the drawing and the matrix. Only the user can settle these.

# 8 Conclusion

The three questions have direct answers.

**Q1.** For a simple undirected graph with adjacency matrix $A$ and a whole number $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length $k$ from $n_i$ to $n_j$ (Theorem 3.5, proved by induction on $k$ from the definition of the matrix product; the statement is standard [2, 3]). It counts walks and not paths: in the board graph $(A^2)_{11} = 1$, the walk $1, 2, 1$, which is neither a path nor a trail (Proposition 3.11).

**Q2.** A graph with $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 4.15, derived here). The proof uses the walk-counting theorem and the fact that every walk contains a path of no greater length, so powers beyond $N - 1$ add nothing; the bound cannot be lowered to $N - 2$ (Proposition 4.23). The board graph, with $S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, is connected; the graph with the edge $n_1n_2$ only, with $S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, is not. Breadth-first search is an alternative that needs no matrix powers [10].

**Q3.** A path between $n_i$ and $n_j$ is a walk from $n_i$ to $n_j$ in which all vertices are pairwise distinct, that is, the textbook's simple path (Definition 1.2.13) [9, p. 22]. The textbook's own word "path" (Definition 1.2.12) allows repeated vertices and is called a trail here. The statement that the course uses the first meaning comes from the assignment text and is an interpretation; the existence of a path between two vertices is the same under both readings (Section 5).

**What remains open.** The reading of the mark above $A$ on the board, the mismatch between the board's drawing and its matrix, and the question which test the lecturer intended are not settled by the sources (Section 7.5).

# A Derivations and checks

This appendix holds the proofs and computations that the main text only states. Every statement here is derived in this paper, except where a citation is given.

## A.1 The distance properties

**Proposition A.1** (Shortest walk, trail, and path agree)**.** *For vertices $u, v$ of a simple graph, the length of a shortest $(u, v)$-walk, of a shortest $(u, v)$-trail, and of a shortest $(u, v)$-path are equal. Each is $+\infty$ when the corresponding set is empty, and the three sets are empty together. So $d(u, v)$ of Definition 2.41 is the same whether "path" in that definition is read as trail or as path.*

*Proof.* Every path is a trail (Proposition 2.35) and every trail is a walk. So the lengths of paths are among the lengths of trails, which are among the lengths of walks. Hence the minimum over walks is at most the minimum over trails, which is at most the minimum over paths. By Lemma 2.45 a shortest walk contains a path of at most the same length, so the minimum over paths is at most the minimum over walks. Therefore the three minima are equal. If there is no walk, there is no trail and no path. If there is a walk, Lemma 2.45 gives a path, so the three sets are empty together. □

Another text makes the same remark in prose: a shortest walk from $u$ to $v$, if it exists, is also a shortest path [3, Section 2.9.4].

**Proposition A.2** (The distance properties)**.** *The three properties of Definition 2.41 follow from the definition.*

*Proof.* Use Proposition A.1 to compute $d$ by shortest walks. (1) Lengths are $0, 1, 2, \ldots$ or $+\infty$, so $d \geq 0$. The walk consisting of $u$ alone has length 0, so $d(u, u) = 0$. If $d(u, v) = 0$ there is a walk of length 0 from $u$ to $v$; it is the single vertex, so $v = u$. (2) If $u_0, \ldots, u_k$ is a $(u, v)$-walk, then $u_k, \ldots, u_0$ is a $(v, u)$-walk of the same length, since each step is an edge and edges are unordered. So the sets of walk lengths from $u$ to $v$ and from $v$ to $u$ are equal, and so are their minima. (3) If $d(u, w)$ or $d(w, v)$ is $+\infty$, the right side is $+\infty$ and the inequality holds. Otherwise take a shortest $(u, w)$-walk and a shortest $(w, v)$-walk and join them at $w$. The result is a $(u, v)$-walk of length $d(u, w) + d(w, v)$, and the shortest $(u, v)$-walk is no longer than this. □

## A.2 The counterexample to "entries count paths"

For the board graph the script `m1_walks.py` lists, for each $k$ from 1 to 5, all walks from $n_1$ to $n_2$ and tests each for distinct vertices (path) and distinct edges (trail). Its output is saved in `appendices/scripts/m1_walks_output.txt`. Table 6 repeats the numbers; the author script `author_numbers.py` recomputes them independently and agrees.

Table 6: Walks from $n_1$ to $n_2$ in the board graph, and how many of them are paths (they are also the trails).

| $k$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $(A^k)_{12}$ = walks | 1 | 0 | 2 | 0 | 4 |
| paths among them | 1 | 0 | 0 | 0 | 0 |

For closed walks at $n_1$ the numbers of walks of length $1, \ldots, 4$ are $0, 1, 0, 2$, and none of them is a path or a trail (script `m1_walks.py`). The general fact is Proposition 3.11.

## A.3 The chain of premises for the connectivity test

The proof of Theorem 4.15 rests on the premises below, each marked with where it comes from.

1. *Walk counting* (Theorem 3.5): the entry $(A^k)_{ij}$ counts walks of length $k$ [2, 3]; proved in Section 3.

2. *Non-negativity* (Corollary 3.9): every entry of every power is a non-negative whole number; derived from item 1.

3. *Walk to path* (Lemma 2.45): a walk contains a path of at most its length [3, Corollary 2.9.7]; the textbook states it for $u \neq v$ [9, p. 23]; proved in Section 2.

4. *Length bound* (Corollary 2.46): a path has at most $N-1$ edges [3, Section 2.9.4]; proved in Section 2.

5. *Path implies trail* (Proposition 2.35): proved in Section 2.

6. *Definition of connected* (Definition 4.1) [9, p. 48].

Forward direction: if $G$ is connected and $i \neq j$, items 6, 3 and 4 give a path of length $k$ with $1 \leq k \leq N-1$, and items 1 and 2 give $(A^k)_{ij} \geq 1$ and hence $S_{ij} \geq 1$. Backward direction: if $S_{ij} > 0$, items 1 and 2 give a walk, and items 3 and 5 give a path in either reading, so $G$ is connected. These are the steps of Proposition 4.13.

## A.4 Proof of the $I + A$ form

*Proof of Proposition 4.14.* Put $m \geq N-1$ and expand $(I+A)^m = (I+A)(I+A)\cdots(I+A)$, with $m$ factors. Each term of the expansion chooses $I$ or $A$ from each factor and multiplies the choices in order. Because $IX = XI = X$ for every matrix $X$, a product with $k$ letters $A$ and $m-k$ letters $I$ equals $A^k$. The number of ways to choose which $k$ of the $m$ factors give $A$ is the binomial coefficient $\binom{m}{k}$. Therefore

$$(I+A)^m = \binom{m}{0} I + \binom{m}{1} A + \binom{m}{2} A^2 + \cdots + \binom{m}{m} A^m.$$

Take an entry $(i, j)$ with $i \neq j$. The identity has 0 there, so

$$\left((I+A)^m\right)_{ij} = \binom{m}{1} (A)_{ij} + \binom{m}{2} (A^2)_{ij} + \cdots + \binom{m}{m} (A^m)_{ij}.$$

Every coefficient $\binom{m}{k}$ with $1 \leq k \leq m$ is a positive whole number, and every $(A^k)_{ij}$ is a non-negative whole number. So the sum is positive if and only if at least one $(A^k)_{ij}$ with $1 \leq k \leq m$ is positive. For $m = N-1$ this is exactly $S_{ij} > 0$. For $m > N-1$, if some $(A^k)_{ij}$ with $k \leq m$ is positive, there is a walk from $n_i$ to $n_j$, hence by Corollary 2.46 a walk of length at most $N-1$, so some $(A^{k'})_{ij}$ with $1 \leq k' \leq N-1$ is positive, that is, $S_{ij} > 0$; the converse is immediate because $N-1 \leq m$. On the diagonal the term $\binom{m}{0} I$ contributes 1 and all other terms are non-negative, so every diagonal entry of $(I+A)^m$ is at least 1 whatever the graph is. □

The idea of the $(I+A)^{N-1}$ form appears in lecture notes in a proof sketch [1]; the expansion above is derived here.

### A.5 Cross-checks by script

All checks are finite tests of statements that are proved in the text.

- *Walk counting.* For every labelled simple graph on $N = 2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs) the script `m1_walks.py` enumerated all walks by depth-first search and compared with $A^k$ for $k = 1, \dots, 5$ when $N \le 4$ and $k = 1, \dots, 3$ when $N = 5$. All 40, 360, 5120 and 76800 tested cases of $(k, i, j)$ agree.
- *Lemma and propositions of Section 2.* The script `m2_walks.py` applied the cutting construction of Lemma 2.45 to all 73 walks of length 0 to 6 in the board graph, all 381 in the triangle and all 4695 in the textbook graph, with 0 failures. On 300 random graphs with $N$ from 2 to 7 (seed 20261027) and 192696 walks of length 0 to 5 it found 0 failures of the lemma, 0 failures of Proposition 2.35, 0 paths longer than $N - 1$ (Corollary 2.46), and 0 disagreements between BFS distance and the shortest walk, trail and path (Proposition A.1).
- *Connectivity test.* The script `m3_connectivity.py` reports in Table 5 the exhaustive agreement with BFS for $N = 2, \dots, 6$, and for the path graph on $N = 3, \dots, 8$ vertices that the smallest $k$ with $(A^k)_{1N} > 0$ is $N - 1$, so that the $(1, N)$ entry of $A + \cdots + A^{N-2}$ is 0 and that of $A + \cdots + A^{N-1}$ is 1 each time.

## B The textbook's terms and numbers

Table 7 lists every textbook definition and result used in the paper, with its printed page, so that each can be checked against the book [9]. Page numbers are the printed numbers of the book; the PDF page is the printed page plus one. The textbook is in Armenian, and its text layer drops most symbols, so formulas were checked against the page images. Only the pages named below were read page by page, and the rest of the book was not read.

Table 7: Textbook definitions and results used in this paper [9].

| Textbook item | Page | Content (English) | Paper |
|---|---|---|---|
| Definition 1.1.1 | 9 | graph $G = (V, E)$, finite $V$, $E$ a set of unordered pairs of distinct vertices | Definition 2.1 |
| Definition 1.1.4 | 11 | adjacent vertices | Definition 2.5 |
| adjacency matrix (after Definition 1.1.6; number not recorded) | 11–12 | $a_{ij} = 1$ if adjacent, else 0; symmetric, zero diagonal; converse | Definition 2.13, Proposition 2.17 |
| Definition 1.2.10 | 21–22 | walk; vertices and edges may repeat; length counts edges with multiplicity | Definition 2.18 |
| Definition 1.2.11 | 22 | closed walk | Definition 2.23 |
| Definition 1.2.12 | 22 | "path": distinct edges (English: trail) | Definition 2.27 |
| Definition 1.2.13 | 22 | "simple path": distinct vertices (English: path) | Definition 2.31 |
| Definition 1.2.14 | 22 | "closed path or cycle": closed trail | Definition 2.36 |
| Definition 1.2.15 | 22 | "simple cycle": only first and last vertex repeat; $C_n$, $n \ge 3$ | Definition 2.37 |
| Definition 1.2.16 | 22–23 | distance | Definition 2.41 |
| Lemma 1.2.1 | 23 | from a walk between different vertices one can extract a simple path | Lemma 2.45 |
| Definition 2.1.1 | 48 | connected graph | Definition 4.1 |
| (no number) | 49 | connected components | Definition 4.5 |
| Remark 2.1.1 | 49 | connected if and only if one component | Remark 4.6 |

**Board and textbook notation.** The board writes $(G, V)$ for the graph, $e_{ij} \in V$ for "the edge $e_{ij}$ belongs to the edge set", $|G| = N$ for the number of vertices, and $n_i, n_j$ for vertices [4, 5]. The textbook writes $G = (V, E)$, $e \in E$, $n = |V|$, and $v_i$ [9]. The paper uses $G = (V, E)$ and $N$ for the number of vertices, and the board's $n_i$ for vertices (Table 2).

## C The whiteboard photographs

Two photographs show the same lecture board at slightly different angles [4, 5]. They are read together. Figure 1 reproduces both.

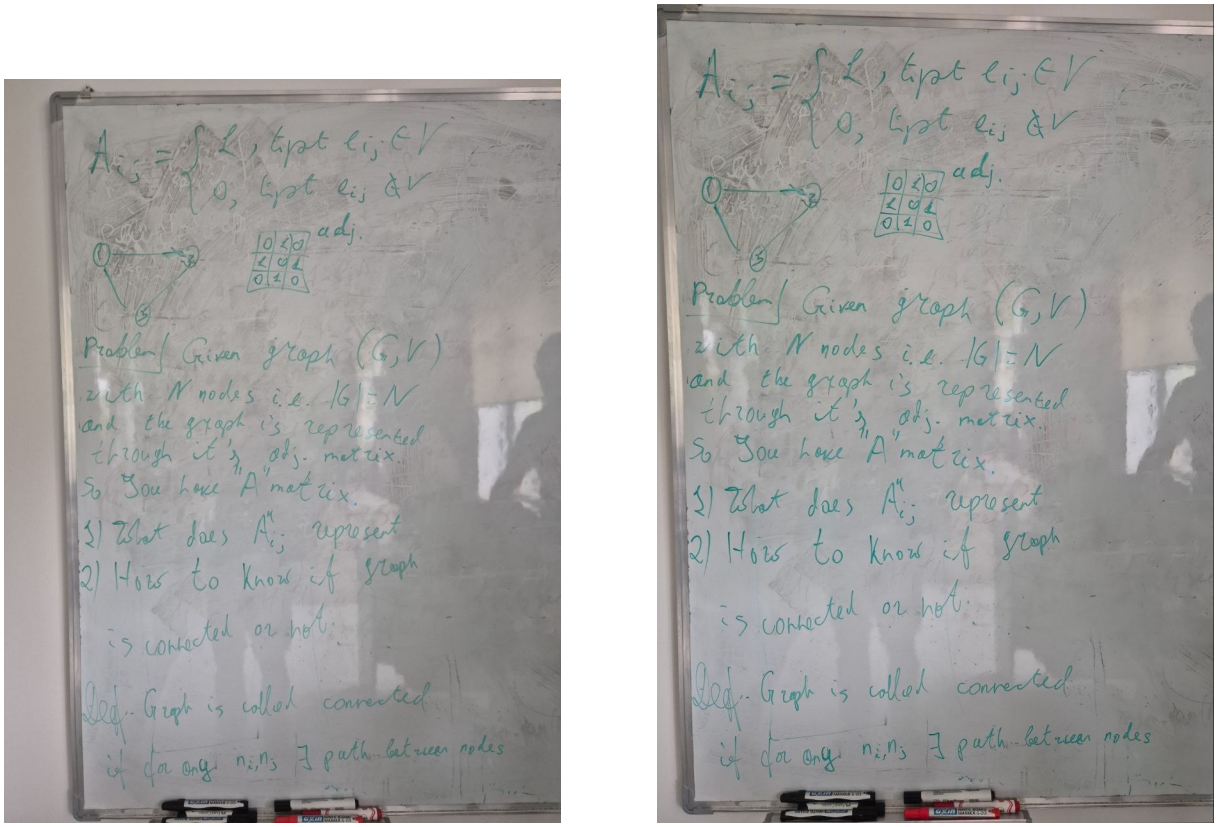



Figure 1: The two photographs of the lecture board [4, 5]. A reflection of a person is visible on the right.

**Transcription in reading order.**

1. $A_{ij} = 1$ if $e_{ij} \in V$, and 0 if $e_{ij} \notin V$; the board's $V$ is the edge set [4, 5].
2. A drawing with vertices labelled $1, 2, 3$, and next to it a matrix labelled "adj." with rows 0 1 0 / 1 0 1 / 0 1 0.
3. "Problem. Given graph $(G, V)$ with $N$ nodes i.e. $|G| = N$ and the graph is represented through it's adj. matrix. So you have $A$ matrix."
4. "1) What does $A_{ij}$ represent", with a small mark above $A$.
5. "2) How to know if graph is connected or not."
6. "Def. Graph is called connected if for any $n_i, n_j$ there exists a path between nodes."

**Clearly visible.** All the prose above, the matrix entries, and the vertex labels.

**Likely interpretation.** The mark above $A$ is an exponent $k$ (interpretation, qualified: the assignment text asks about $A^k$, but in both photographs the mark looks like a small $n$ or a pair of quotation marks). The arrowhead on the line 1–2 is a stray mark, because the matrix is symmetric and the problem is about undirected graphs (interpretation).

**Unclear or in conflict.** The drawing shows a line from 1 to 2 with an arrowhead at 2, a line from 1 to 3, and a line from 2 to 3, which would be three edges, while the matrix encodes only the edges 1–2 and 2–3 [4, 5]. Edge by edge: the edge 1–2 is in both; the edge 2–3 is in both; the edge 1–3 is in the drawing only. The paper follows the matrix and reports the mismatch.

# D Reproducibility

Every number in the paper that is a matrix entry, a sum, or a count was computed by one of the scripts below. All are pure Python with integer arithmetic, so no rounding occurs. Run each one from the paper folder with `python3 -I`; the saved outputs sit beside the scripts.

`appendices/scripts/author_numbers.py` computes the matrices of Section 6 and writes them to `figures/generated_numbers.tex`, which the paper inputs; it also checks each power against a direct walk enumeration, and computes the walk and path counts of Table 6 and the cost counts of Section 7.4. Output: `author_numbers_output.txt`.

`appendices/scripts/m1_walks.py` board graph powers, explicit walks, walks against paths, brute-force check of Theorem 3.5. Output: `m1_walks_output.txt`.

`appendices/scripts/m2_walks.py` classification of sequences as walk, trail, path, cycle; the cutting construction of Lemma 2.45; Table 4. Output: `m2_walks_output.txt`.

`appendices/scripts/m3_connectivity.py` the tests of Section 4, the examples, the sharpness table, the exhaustive check of Table 5, and the cost counts. It takes about thirty seconds. Output: `m3_connectivity.out.txt`.

The scripts are copies of the ones in the folder `modules/`, which are kept as the module records.

# References


[1] Shuliang Bai. *Important Properties and Applications of the Adjacency Matrix (20250310).* Lecture notes, BIMSA, dated 2025-03-10 in the title. 2025. URL: `https://bimsa.net/doc/notes/48535.pdf?id=0.3935315718408674` (visited on Oct. 27, 2026).

[2] *Chapter 1 Matrices and Graphs, 1.1 The Adjacency Matrix (linear algebra course notes, NUI Galway).* Lecture notes, School of Mathematical and Statistical Sciences, University of Galway; author and date do not appear in the file. n.d. URL: `https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf` (visited on Oct. 27, 2026).

[3] Darij Grinberg. *An introduction to graph theory (Text for Math 530 in Spring 2022 at Drexel University).* arXiv:2308.04512v3 [math.HO], Spring 2025 edition, 7 June 2025; lecture-note text, 454 pp.; page locators are the document's own. 2025. URL: `https://arxiv.org/abs/2308.04512` (visited on Oct. 27, 2026).

[4] *Lecture whiteboard photo 1.* File `sources/whiteboard-1.jpg`. Supplied by the user. Photograph of a lecture board; author and date unknown. Accessed 2026-10-27. n.d.

[5] *Lecture whiteboard photo 2*. File `sources/whiteboard-2.jpg`. Supplied by the user. Second photograph of the same board; author and date of the board unknown. Accessed 2026-10-27. n.d.

[6] Eric Lehman, F. Thomson Leighton, and Albert R. Meyer. *9.3: Adjacency Matrices (Mathematics for Computer Science; MIT OpenCourseWare), Engineering LibreTexts*. Engineering LibreTexts, CC BY-NC-SA; page last modified 2021-07-02; publication year of the underlying book is not on the page. n.d. URL: `https://eng.libretexts.org/Bookshelves/Computer_Science/Programming_and_Computation_Fundamentals/Mathematics_for_Computer_Science_(Lehman_Leighton_and_Meyer)/02%3A_Structures/09%3A_Directed_graphs_and_Partial_Orders/9.03%3A_Adjacency_Matrices` (visited on Oct. 27, 2026).

[7] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *JSc EUSL (Journal of Science, Eastern University, Sri Lanka)* 12.1 (2021), pp. 13–21. DOI: `10.4038/jsc.v12i1.29`. URL: `https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf` (visited on Oct. 27, 2026).

[8] *Path (graph theory)*. Wikipedia, English edition, revision of 5 June 2026 (oldid 1357845833) as fetched. n.d. URL: `https://en.wikipedia.org/wiki/Path_(graph_theory)` (visited on Oct. 27, 2026).

[9] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Grafneri tesutyun (Graph Theory: a study guide)*. File `Grafneri_tesutyun.pdf`, course textbook in Armenian. Supplied by the user. Yerevan State University Press, Yerevan. ISBN 978-5-8084-1995-7. Locators are printed page numbers. 2015.

[10] Éric Schost. *CS 341: Algorithms, Lecture 10: Graphs, breadth first search*. Lecture slides, David R. Cheriton School of Computer Science, University of Waterloo, Winter 2026. 2026. URL: `https://www.student.cs.uwaterloo.ca/~cs341/lectures/lecture10.pdf` (visited on Oct. 27, 2026).

[11] Eric W. Weisstein. *Connected Graph*. Wolfram MathWorld; no publication date on the page. n.d. URL: `https://mathworld.wolfram.com/ConnectedGraph.html` (visited on Oct. 27, 2026).